

Project 1: Clinical Decision Support System with RAG and KG

Concept

Develop a system that augments a Large Language Model (LLM) with a comprehensive knowledge base of medical literature, clinical guidelines, and anonymized Electronic Health Records (EHRs). The system should assist clinicians in diagnosis, treatment planning, and understanding rare conditions by providing evidence-based, contextually relevant information.

Healthcare Value:

- **Improved Diagnosis:** Help clinicians with difficult or rare diagnoses by rapidly sifting through vast medical literature and patient histories.
- **Personalized Treatment:** Support tailored treatment plans based on a patient's unique profile, medical history, and current best practices.
- **Reduced Information Overload:** Provide concise, evidence-backed answers, saving clinicians time.

Key Features for Demo:

- **Contextual Question Answering:** Ask questions like, "What are the common comorbidities for Type 2 Diabetes in patients over 60 with hypertension, and what are the recommended first-line treatments based on the latest guidelines?"
- **Source Citation and Explainability:** The RAG system must cite the source documents (research papers, clinical guidelines, EHR notes) for its answers, demonstrating transparency and trustworthiness.
- **"What If" Scenarios:** Allow clinicians to explore different treatment pathways or diagnostic hypotheses by asking "What if we consider X treatment given Y patient condition?"
- **Pattern Discovery (RAG-enhanced):** Identify subtle relationships between medications, side effects, and patient demographics across a large corpus of notes that might not be explicitly stated in a single document. For example, discover an emerging trend of a particular adverse drug reaction (ADR) associated with a new combination of drugs, even if individual studies only mention single drug ADRs.

Demonstrate following capabilities:

1. Identifying Potential Drug-Induced Symptoms

Question: “Identify all patients who reported symptoms like 'dizziness' or 'fatigue' in their free-text clinical notes, and who are currently on 'Lisinopril'. For these patients, list their initial diagnoses and the full content of the note where the symptom was mentioned, so a clinician can review if it's a potential side effect.”

Suggested approach: Hybrid: Vector DB for Symptom Extraction + Graph for Medication Link)

Hybrid Benefit:

- *Vector DB (RAG):* Would be used to semantically search the unstructured `clinical_notes.content` for mentions of "dizziness" or "fatigue" (and synonyms/related phrases), allowing flexible natural language queries for symptoms.
- *Graph DB:* Once patients with relevant symptoms are identified, the Graph DB would efficiently link these patients to their `medications_on_record` (specifically "Lisinopril") and `initial_diagnoses`, providing structured context.
- *Nested/Discovery:* Discovers potential drug-symptom correlations by connecting unstructured symptom data (from notes) to structured medication data. It allows "discovery" of potential adverse drug reactions.

2: Tracking Symptom Evolution in a Specific Patient Cohort

Question: “For all male patients over 60 years old who have an initial diagnosis of 'Hypertension', trace their `clinical_notes` chronologically. Are there any patients in this group who initially reported 'dizziness' but later reported 'headaches' after being prescribed 'Lisinopril'? If so, list their patient ID and the dates of these specific symptom mentions.”

Suggested approach:

Graph DB - Nested/Discovery & Temporal Analysis)

Graph DB Benefit: Efficiently filters patients by multiple structured attributes (gender, age, `initial_diagnoses`). It then allows traversal through nested `clinical_notes` based on date to detect a sequence or evolution of symptoms.

Nested/Discovery: Demonstrates navigating deeply nested data (`clinical_notes` array, then date and content within each note) and discovering temporal patterns in symptom presentation related to a specific diagnosis and medication.

3. Finding Patients with Co-occurring Conditions and Specific Complaint Patterns

Question: "Find patients who have both 'Type 2 Diabetes' and 'Hypertension' as initial diagnoses. Among these, identify anyone whose chief complaint in *any* clinical note semantically matches 'unusual thirst and frequent urination'. For such patients, list their medications_on_record and the date of the relevant note."

Suggested Approach

Hybrid: Vector DB for Chief Complaint + Graph for Multi-attribute Filtering

Hybrid Benefit:

- Vector DB (RAG): Semantic search on the unstructured clinical_notes.content (specifically extracting the Chief complaint: part) for the phrase "unusual thirst and frequent urination," finding patients even if the exact wording varies.
- Graph DB: Efficiently identifies patients based on the structured initial_diagnoses ('Type 2 Diabetes', 'Hypertension') and then links to the medications_on_record for contextual information.

Nested/Discovery

Combines precise structured filtering with flexible unstructured content search to identify a highly specific patient cohort and relevant clinical context for decision support.

4. Identifying Patients with Recurring Symptoms on Specific Medications "Which medications on record are most frequently associated with patients reporting 'fatigue' in their clinical notes on *multiple distinct dates*? For the top 3 such medications, list the patient IDs and dates of notes where 'fatigue' was reported, illustrating recurring symptom patterns."

Suggested Approach:

Graph DB - Nested & Aggregation

Graph DB Benefit: This requires iterating through patients, then their notes, identifying symptom mentions, linking back to medications, and then performing aggregation (counting distinct symptom mentions per patient per medication) to find recurring patterns.

Nested/Discovery: Explores recurring patterns of symptoms within nested clinical_notes and correlates them with medications_on_record, allowing the "discovery" of potential drug-symptom associations based on frequency.

5. Providing Contextual Information for a Patient with New Symptoms

A patient (e.g., Richard Frank, P002) presents with a new complaint of 'sudden blurry vision'

Help the doctor by

- a) providing summary history of Richard Frank, any earlier diagnosis such as high BP/diabetes
- b) searching clinical notes for blurry vision or semantically similar terms and what other patients with this symptom were diagnosed with. Print the name, symptom, diagnosis and date diagnosed for those patients

Data Sources:

Generate synthetic dataset that contains the patterns to meaningfully answer the questions/address the scenarios below

Tip: copy paste the entire text above into your favourite AI chatbot such as Claude4 and ask it to create python script to generate the dataset that exhibits the patterns needed to answer above questions. Ask it to also validate the patterns in the script.

A reference dataset is provided for you in the project1-dataset folder

The provided **synthetic dataset** contains:

100 patient records and saved to synthetic_patient_ehr.json

30 medical documents and saved to synthetic_medical_knowledge.json

Patterns are verified in the data:

--- Subtle Drug Interaction Pattern ---

Patient P000 (Christopher Chan) on DrugA and DrugB shows 'unusual fatigue' in note from 2024-01-09.

--- T2D + Hypertension Comorbidity Pattern ---

Patient P000 (Christopher Chan) with T2D and Hypertension shows comorbidity mention in note from 2024-09-14.

DEMO SCRIPT

See below for additional demo scenarios and question+answer pairs :

<https://chatgpt.com/share/68d2619d-3268-800f-b220-4d5ab0176d3f>

reproduced below:

1) Multi-Hop: Drug Interaction → Symptom (“fatigue”)

Demo question:

Which patients were on >1 medication (e.g., DrugA + DrugB) and later reported “fatigue” in their notes? Show patient IDs, drugs, and note dates.

Expected output (sample)

patient_id	name	meds_at_time	note_date	note_snippet	evidence
P000	Christopher Chan	DrugA, DrugB	2024-01-09	“...reports unusual fatigue over past week; denies fever...”	[EHR:p000_note_2024-01-09]
P017	Maria Gomez	DrugA, Metformin	2024-03-22	“...persistent fatigue ; considering med interaction eval...”	[EHR:p017_note_2024-03-22]
P031	Liam Patel	DrugB, Lisinopril	2024-02-11	“... fatigue & lightheadedness since dose increase...”	[EHR:p031_note_2024-02-11]

Why it “wows”: Crosses **(Patient → Meds → Notes)** and filters for **polypharmacy + symptom emergence**.

2) Temporal Evolution: Dizziness → Headaches on Lisinopril (Male > 60, HTN)

Demo question:

Trace male patients >60 with Hypertension. Did any first report “dizziness” and later “headaches” while on Lisinopril? List patient IDs and dates.

Expected output (sample)

patient_id	age	sequence_detected	dizziness_date	headaches_date	med	evidence
P024	68	Yes	2024-04-05	2024-05-02	Lisinopril	[EHR:p024_note_2024-04-05], [EHR:p024_note_2024-05-02]
P039	71	Yes	2024-02-18	2024-03-01	Lisinopril	[EHR:p039_note_2024-02-18], [EHR:p039_note_2024-03-01]

Inline snippets

- P024 (2024-04-05): “new **dizziness** when standing...”
- P024 (2024-05-02): “recurrent **headaches**, BP controlled...”

Why it “wows”: Shows **longitudinal traversal** and **ordered symptom transitions** gated by structured filters.

3) Co-Occurring Dx + Semantic Complaint (T2D n HTN → “thirst & frequent urination”)

Demo question:

Find patients with both T2D and Hypertension. Among them, who had a chief complaint semantically matching “unusual thirst and frequent urination”? Return medications and note dates.

Expected output (sample)

patient_id	name	comorbidities	meds_on_record	note_date	matched_complaint_snippet	evidence
P000	Christopher Chan	T2D, Hypertension	Metformin, Lisinopril, DrugA, DrugB	2024-09-14	“ excessive thirst, polyuria esp. at night”	[EHR:p000_note_2024-09-14]
P046	Aisha Khan	T2D, Hypertension	Metformin, Amlodipine	2024-06-21	“drinking more water, frequent urination ”	[EHR:p046_note_2024-06-21]

Why it “wows”: **KG** does co-morbidity filtering; **RAG** semantically normalizes complaint phrasing.

4) Recurring Symptom Aggregation (Top Meds ↔ Multi-date “fatigue”)

Demo question:

Which medications are most frequently associated with patients reporting “fatigue” across multiple distinct visits? Show top 3 meds, patient IDs, and note dates.

Expected output (sample)

Top meds by #patients with ≥2 fatigue-notes

1. **Lisinopril** — 7 patients

- P031: 2024-02-11, 2024-03-03 ([EHR:p031_note_2024-02-11], [EHR:p031_note_2024-03-03])
- P058: 2024-01-27, 2024-02-09 ([EHR:p058_note_2024-01-27], [EHR:p058_note_2024-02-09])
- ...

2. **DrugA** — 5 patients

- P000: 2024-01-09, 2024-02-02 ([EHR:p000_note_2024-01-09], [EHR:p000_note_2024-02-02])
- P017: 2024-03-22, 2024-04-13 ([EHR:p017_note_2024-03-22], [EHR:p017_note_2024-04-13])

3. **DrugB** — 4 patients

- P000: 2024-01-09, 2024-02-02 (same notes as above if both meds present)
- P041: 2024-05-06, 2024-05-28 ([EHR:p041_note_2024-05-06], [EHR:p041_note_2024-05-28])

Why it “wows”: Graph aggregations reveal drug-symptom **frequency patterns** across the population.

5) Patient-Centric Narrative: Richard Frank (P002) → New “blurry vision”

Demo question:

Richard Frank presents with ‘sudden blurry vision’. Summarize his history; then compare with others who had blurry vision (their symptom → diagnosis → date).

Expected output (sample)

a) Richard Frank (P002) — Summary

- **History:** Hypertension (dx 2021-08-10), Prediabetes (2023-02-19)
- **Meds:** Amlodipine (current), Metformin (started 2024-07-01)
- **New complaint** (2024-09-20): “**sudden blurry vision** since morning”
- **Vitals/Labs context:** BP mildly elevated; last HbA1c 6.7% (2024-08-12)
Evidence: [EHR:p002_summary], [EHR:p002_note_2024-09-20]

b) Similar-symptom cohort (blurry vision → dx)

patient_id	name	note_date	symptom_snippet	subsequent_dx (and date)	evidence
P019	Evelyn Brooks	2024-06-03	“episodes of blurry vision late afternoon”	T2D progression (2024-07-15)	[EHR:p019_note_2024-06-03], [EHR:p019_dx_2024-07-15]
P057	Arjun Iyer	2024-05-11	“ blurry vision with headaches”	Migraine (2024-05-29)	[EHR:p057_note_2024-05-11], [EHR:p057_dx_2024-05-29]
P072	Hannah Lee	2024-08-19	“transient visual blur post-exercise”	Dehydration (2024-08-20)	[EHR:p072_note_2024-08-19], [EHR:p072_dx_2024-08-20]

Guidance to clinician (auto-generated)

- Check R/O **hyperglycemia**-related visual changes vs. ocular pathology; consider same-day glucose, ophthalmology referral.
- Compare Richard’s profile to P019 (T2D progression pattern).
References: [MEDGUIDE:ophthalmic_changes_T2D_2023], [MEDGUIDE:HTN_eye_findings_2024]

6) Emerging Pattern Discovery: Polypharmacy → Unusual ADR clusters

Demo question:

Are there clusters of patients showing unusual side effects when on multiple meds (DrugA + DrugB, or Lisinopril + another drug)? Show clusters and patient IDs.

Expected output (sample)

Cluster A (DrugA + DrugB)

- **Key symptoms:** **unusual fatigue**, lightheadedness
- **Patients:** P000 (Christopher Chan), P041, P055
- **Representative snippets:**
 - P000 (2024-01-09): “**unusual fatigue**; denies infection”
[EHR:p000_note_2024-01-09]

- P041 (2024-05-06): “fatigue, reduced activity tolerance”
[EHR:p041_note_2024-05-06]

Cluster B (Lisinopril + diuretic)

- **Key symptoms:** dizziness → headaches sequence, occasional cramps
- **Patients:** P024, P039, P063
- **Snippets:**
 - P024 (2024-04-05 → 2024-05-02) progression documented
[EHR:p024_note_2024-04-05], [EHR:p024_note_2024-05-02]

Why it “wows”: Finds **non-obvious groupings** across meds & timelines (not in any single document).

7) Guidelines + EHR Cross-Check (RAG + KG)

Demo question:

According to latest guidelines, what is first-line therapy for T2D with Hypertension? Do our patient records (e.g., Christopher Chan) follow this?

Expected output (sample)

Guideline summary (RAG)

- **Recommendation:** For **T2D + Hypertension**, initiate **Metformin** and consider **ACE inhibitor** for BP control; reinforce lifestyle modifications.
- **Source:** “[Type 2 Diabetes with Hypertension: Initial Therapy 2024]” →
[MEDGUIDE:t2d_htn_2024_guideline_doc_07]
- **Citation handle:** [MEDGUIDE:t2d_htn_2024_guideline_doc_07 §3.1–3.3]

EHR Cross-check (KG)

patient_id name		dx_combo	meds_on_record	guideline_alignment
P000	Christopher Chan	T2D + HTN	Metformin, Lisinopril , DrugA, DrugB	Aligned (Metformin + ACE-i)
P046	Aisha Khan	T2D + HTN	Metformin, Amlodipine	Partial (Metformin ok; consider ACE-i)

Why it “wows”: Trust via **transparent citations** + immediate **compliance check** inside your EHR graph.

Evidence Handle Conventions (so your UI can link them)

- **EHR:** [EHR:p<id>_note_<YYYY-MM-DD>], [EHR:p<id>_dx_<YYYY-MM-DD>], [EHR:p<id>_summary] → resolve into synthetic_patient_ehr.json entries (patient_id, note_id/date).
- **Guidelines / Knowledge:** [MEDGUIDE:<doc_key>] → resolve into synthetic_medical_knowledge.json (title, year, section indices).